

Logistic Regression

Why use logistic regression?

- There are many important research topics for which the dependent variable is "limited."
- For example: voting, morbidity or mortality, and participation data is not continuous or distributed normally.
- Binary logistic regression is a type of regression analysis where the **dependent variable is a dummy variable: coded 0 (did not vote) or 1(did vote)**

The Linear Probability Model

In the OLS regression:

$$Y = a + bX + e ; \text{ where } Y = (0, 1)$$

1. The error terms are **heteroskedastic**
2. e is not **normally distributed** because Y takes on only two values
3. **The predicted probabilities can be greater than 1 or less than 0**

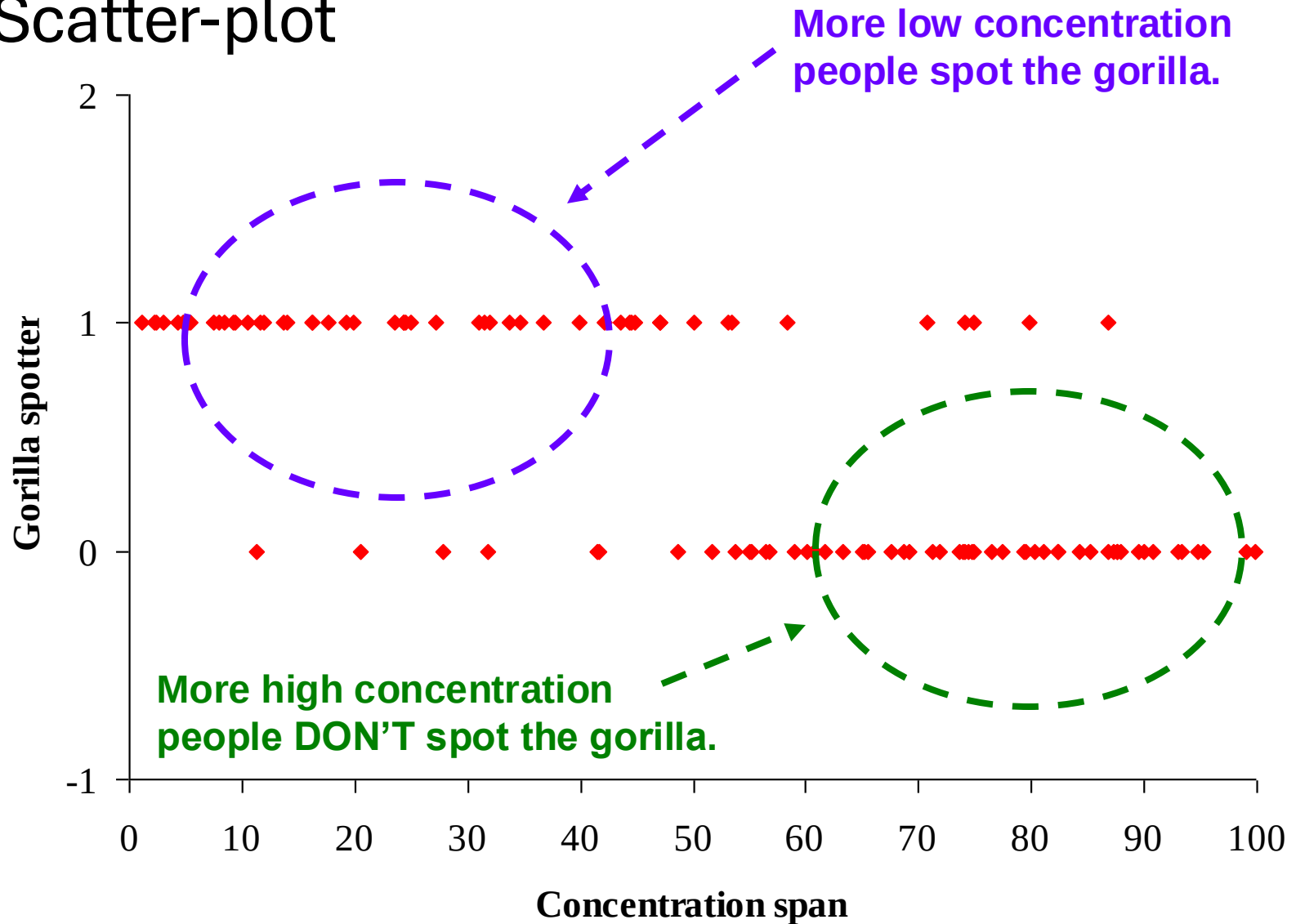
Example: Predicting gorilla sightings

- To predict whether the gorilla will be spotted by a person with a particular set of characteristics.
 - In this case, let's say with a particular concentration span (measured on a 1-100 scale).
- So, what we want to know is the probability that any person will be a gorilla spotter or not for any value of concentration span.
- We could use simple linear regression (SLR) here, with the dependent variable coded as 0 (no gorilla spot) or 1 (gorilla spotted).

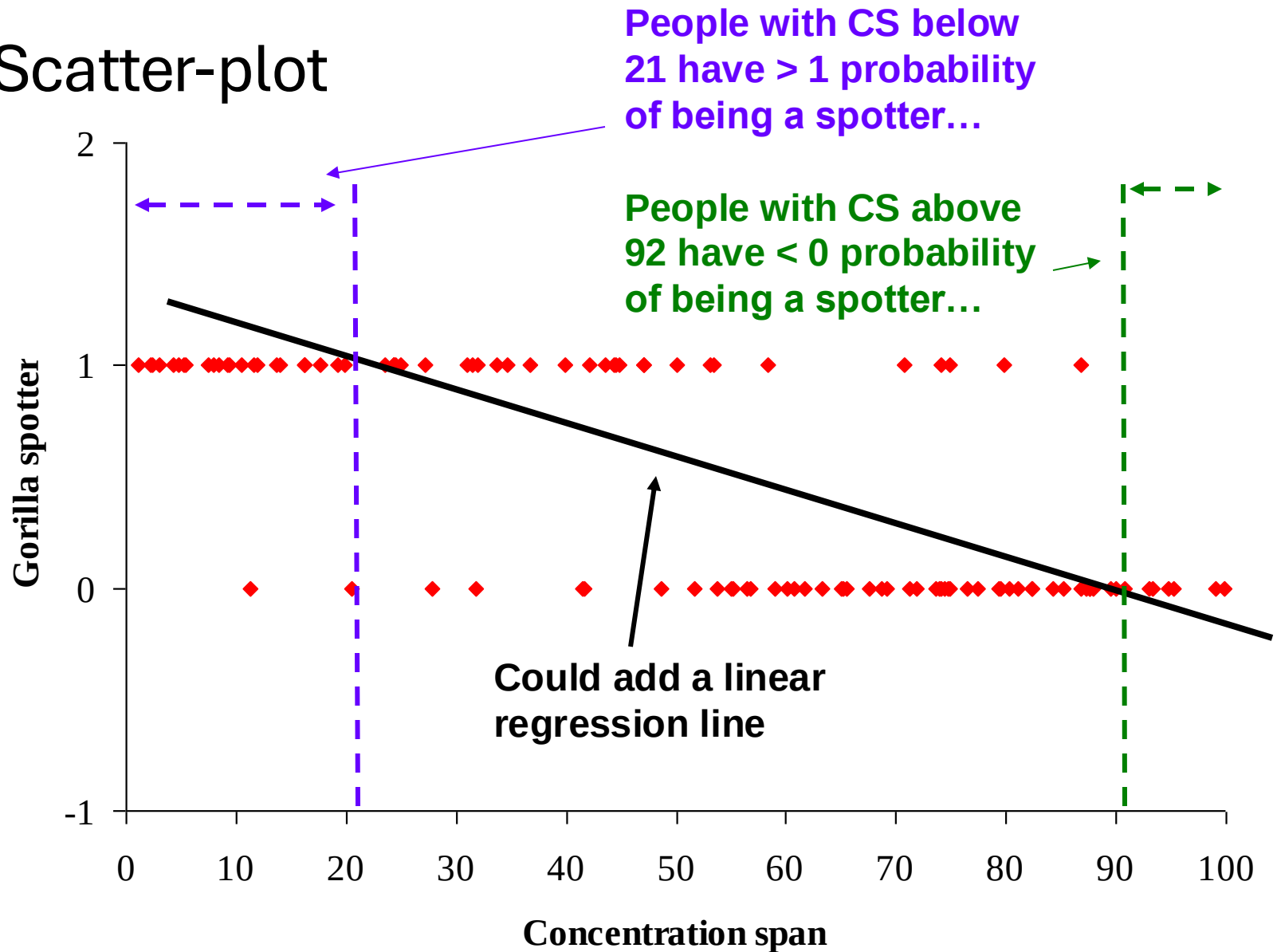
What's wrong with SLR?

- We want to predict a probability, this can only vary between zero and 1.
- But our SLR may predict values that are below zero or above 1...

Scatter-plot



Scatter-plot



The Logistic Regression Model

The "logit" model solves these problems:

$$\ln[p/(1-p)] = \alpha + \beta X + e$$

- p is the probability that the event Y occurs, $p(Y=1)$
- $p/(1-p)$ is the "**odds ratio**"
- $\ln[p/(1-p)]$ is the **log odds ratio**, or "**logit**"

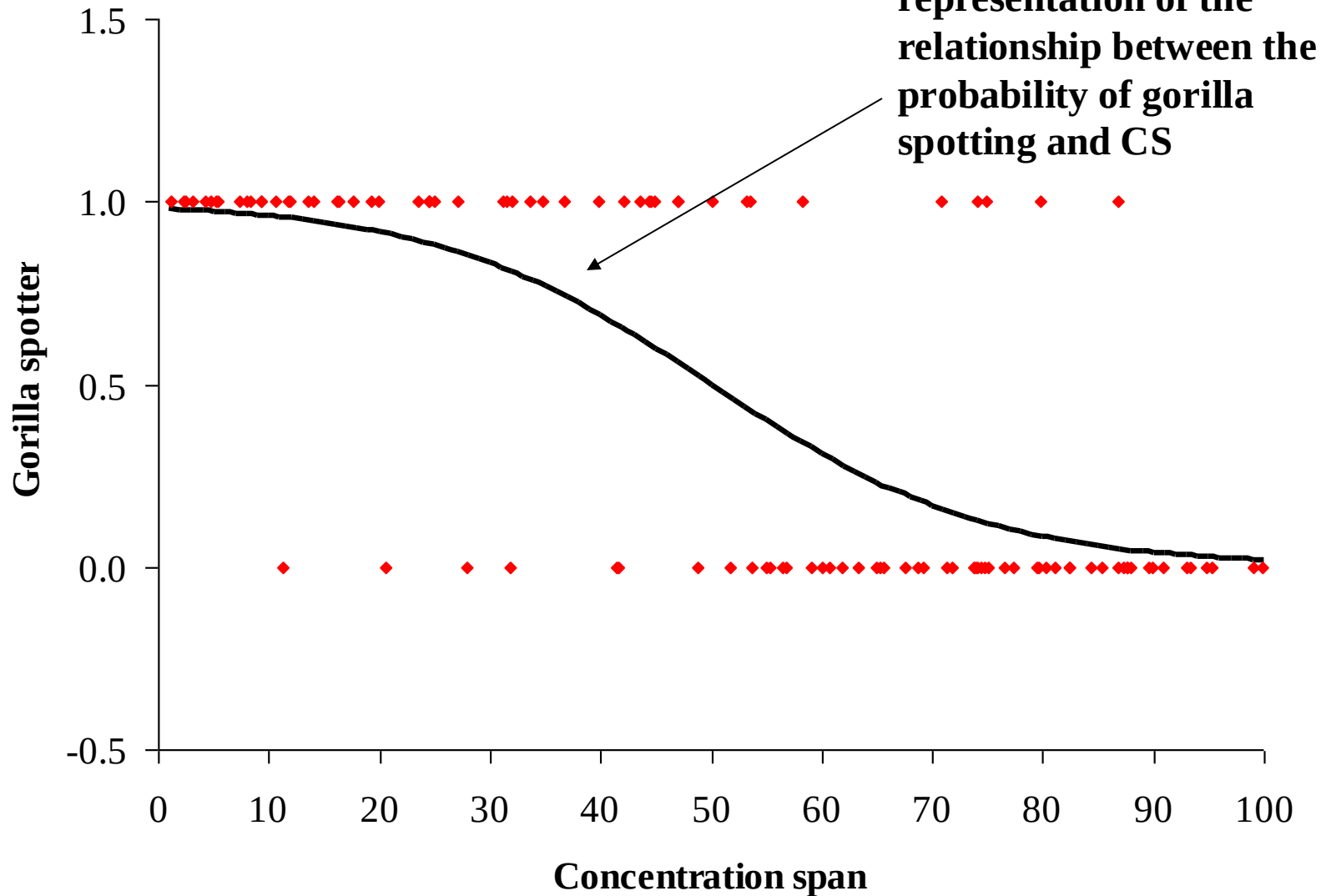
The Logistic Regression Model

1. The logistic distribution constrains the estimated probabilities to lie between 0 and 1.
2. The estimated probability is:

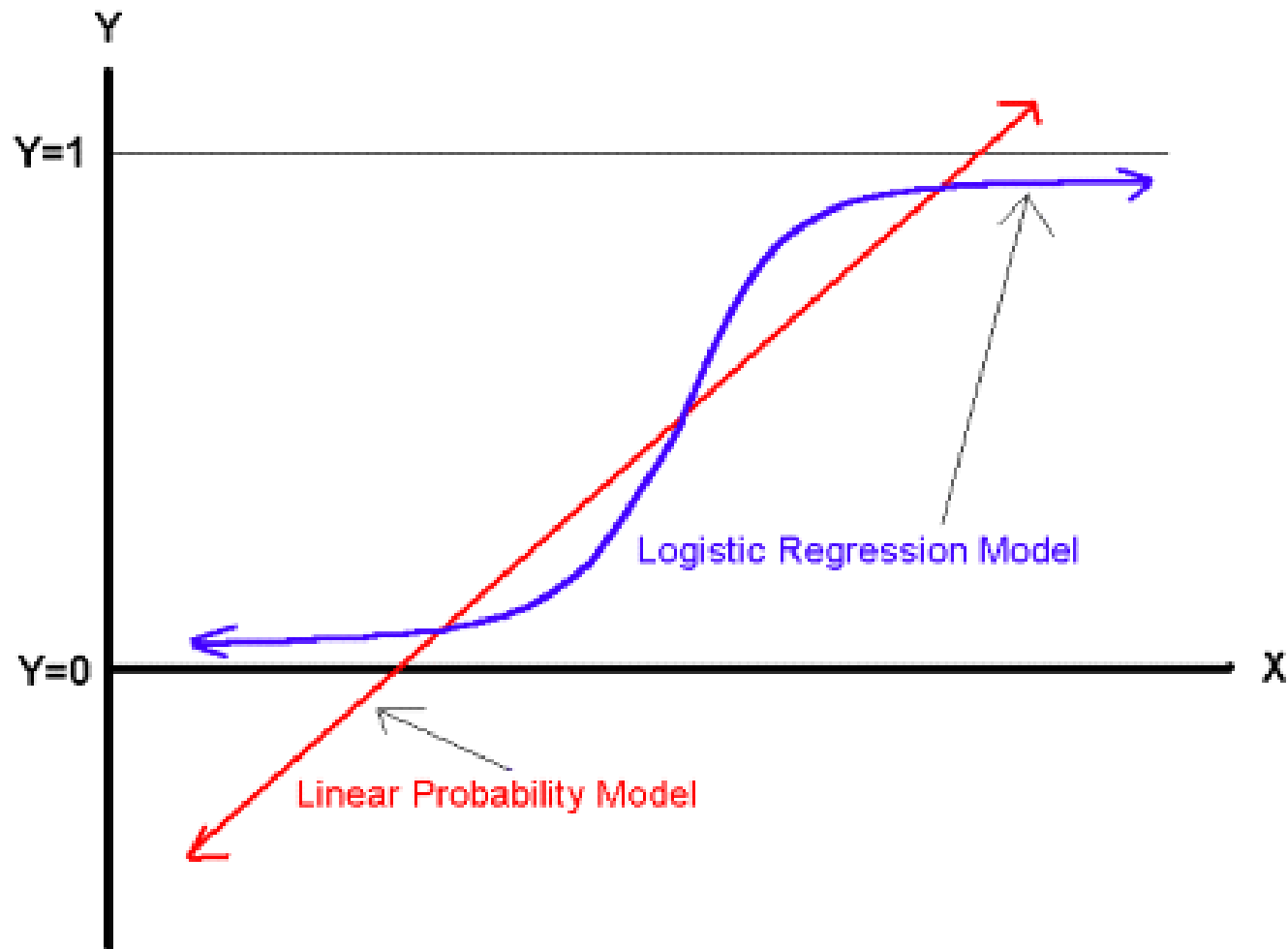
$$p = 1/[1 + \exp(-\alpha - \beta X)]$$

3. if you let $\alpha + \beta X = 0$, then $p = .50$
4. as $\alpha + \beta X$ gets really big, p approaches 1
5. as $\alpha + \beta X$ gets really small, p approaches 0

Scatter-plot and Logistic R



Comparing the LP and Logit Models



The logistic transformation

- This type of relationship is described by a special formula.
 - Remember, if the relationship was linear then the equation is just:

$$\pi = \alpha + \beta X$$

- But the relationship on the graph is actually described by:

$$\log\left(\frac{\pi}{1-\pi}\right) = \alpha + \beta X$$

The logistic transformation (2)

$$\log\left(\frac{\pi}{1-\pi}\right) = \alpha + \beta X$$

This is just the *odds*.

As the probability increases (from zero to 1), the odds increase from 0 to infinity.

The log of the odds then increases from $-\infty$ to $+\infty$.

So if β is 'large' then as X increases the log of the odds will increase steeply.

The steepness of the curve will therefore increase as β gets bigger.

Fitting this model

- ***Maximum likelihood*** is a statistical method for estimating the coefficients of a model. It is an *iterative process* that estimates the best fitted equation.
- The likelihood function (L) measures the probability of observing the particular set of dependent variable values ($p_1, p_2, \dots, p_n$) that occur in the sample:
$$L = \text{Prob} (p_1 * p_2 * * * p_n)$$
- The higher the L, the higher the probability of observing the ps in the sample.

Fitting this model

- MLE involves finding the coefficients (α, β) that makes the log of the likelihood function ($LL < 0$) as large as possible
- Or, finds the coefficients that make -2 times the log of the likelihood function ($-2LL$) as small as possible
- The maximum likelihood estimates solve the following condition:
$$\{Y - p(Y=1)\}X_i = 0$$
summed over all observations, $i = 1, \dots, n$

Interpreting Coefficients

- Since:

$$\ln[p/(1-p)] = \alpha + \beta X + e$$

- The slope coefficient (β) is interpreted as the rate of change in the "log odds" as X changes.

Interpreting Coefficients

- So an increase in X of 1 unit will decrease our log (odds) by 0.07.
- If we *antilog* both sides then we could see how the odds change.

$$\log\left(\frac{\hat{\pi}}{1 - \hat{\pi}}\right) = a + bX$$

$$\log\left(\frac{\hat{\pi}}{1 - \hat{\pi}}\right) = 3.69 - 0.07X$$



**Remember the ‘hat’
sign means the
predicted value.**

Interpreting Coefficients

- Antilog both sides and we get the odds on the LH side.
- If we enter a value of X we can work out what the predicted odds will be.
- Thus the odds of spotting the gorilla (as opposed to not spotting the gorilla) are nearly 5. For every 5 spotters there should be one non-spotter.

$$\frac{\hat{\pi}}{1 - \hat{\pi}} = e^{a+bX}$$

$$\frac{\hat{\pi}}{1 - \hat{\pi}} = e^{3.69 - 0.07X}$$

$$\frac{\hat{\pi}}{1 - \hat{\pi}} = e^{3.69 - 0.07X}$$

$$\frac{\hat{\pi}}{1 - \hat{\pi}} = e^{3.69 - 0.07 \times 30} = 4.90$$

Interpreting Coefficients

- We can also think about what happens to the odds when we increase X by a certain amount.
- Another way of writing e^{a+bX} is $e^a(e^b)^X$. That means that **a one unit increase in X multiplies the odds by e^b** (as it's to the power of 1).
- In our case therefore a one unit increase in X multiplies the odds by $e^{-0.07}$, or 0.93.
 - When X increases from 30 to 31, the odds are 4.90×0.93 , or 4.56.
 - When X increases from 30 to 40, the odds are $4.90 \times (0.93)^{10}$, or 2.37.

Adding extra variables

- Including other interval level independent variables and categorical independent variables is as easy as in multiple linear regression.

Comparing models

- One of the most important differences between logistic regression and linear regression is in how we compare models.
 - Remember for linear regression we looked at how the **adjusted R^2** changed. If there was a significant increase when we added another variable (or interaction) then we thought the model had improved.
- For logistic regression there are a variety of ways of looking model improvement.

Comparing models

- The best way of comparing models is to use something called the ***likelihood-ratio test***.
- When we were using OLS regression, we were trying to minimize the sum of squares, for logistic regression we are trying to maximize something called the *likelihood function* (normally called L).
- To see whether our model has improved by adding a variable (or interaction, or squared term), we can compare the **maximum of the likelihood function** for each model (just like we compared the R^2 before for OLS regressions).

Comparing models

- In fact, just to complicate matters we actually compare the maximized values of $-2 \times \log L$.

$$LR = (-2 \times \log L_o) - (-2 \times \log L_1)$$

First model's maximised value

Second model's maximised value

- By logging the L s and multiplying them by -2, this statistic conveniently ends up with a chi-square distribution. This means we test whether there is a statistically significant improvement with reference to the χ^2 distribution.